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Weak Boson Processes

Responsible:

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Public mailing list:*

atlas-phys-pmg-weak-boson@cern.ch

***Regular meetings:** Tuesday, biweekly, 16:00-17:30 CERN time: <https://indico.cern.ch/category/10312/>

Introduction

The PMG Weak Boson Processes subgroup is responsible for physics modelling aspects of the following processes:

- Weak bosons (V+jets, V=W, Z)
- Multiple bosons (VV, VVV, at least one V=W, Z)
 - Note:** single- or multiple-photon processes are under the scope of the [PmgJetAndPhotonProcesses](#)
- Single and Di-Higgs samples (in collaboration with the HIGP PMG contacts). A dedicated summary page of the Higgs samples can be found in [HiggsDiHiggsMCSamples](#)

Sample overview

List of samples (NEW)

Please use the central page to find the list of the latest available MC samples. The list below is incomplete and will be removed soon.

MC16 samples: [CentralMC16ProductionListNew](#)

MC20 samples: [CentralMC20ProductionListNew](#)

MC21 samples: [CentralMC21ProductionListNew](#)

MC23 samples: [CentralMC23ProductionListNew](#)

General recipes

Here are some general recipes on which samples to use and how to use them. Given the complexity of Weak Boson processes (production, decay, multiplicity etc) and the phase space you are interested in, there are no universal recipes that work for everyone. Feel free to contact via the mailing lists above if you have any questions.

- **Which samples/generators to use?** For most commonly used weak boson processes, at least two MC generators are used to produce official samples: multi-leg merged/matched samples (like Sherpa, [MadGraph_aMC@NLO+Pythia8/Herwig](#)) and inclusive samples (like [PowhegBox](#), Herwig, VBFNLO, etc). If you are looking at a phase space that is involved with high jet multiplicity, it's recommended to use the multi-leg MC samples.
- **Which processes to use?** As for production mechanism, single-boson and diboson processes usually involve various production modes: quark-initiated, pure electroweak production (vector-boson-fusion or vector-boson-scattering), and gluon-initiated (for diboson). Therefore, all different production modes should be considered, and the relative contribution for each production mode could be quite different depending on the phase space. Care must be given to cases where Higgs contribution should be included, since there could be overlap between weak boson samples and dedicated Higgs samples (for example, electroweak VVjj samples also include VBF Higgs contribution). For detailed sample overlap, please refer to [PmgWeakBosonProcesses#Overlap_with_Higgs_signal_sample](#) and the detailed description for each process below.

As for decay final states, one should also be careful when dealing with on-shell and off-shell contributions, since overlap could also happen between samples (for example, electroweak VVjj samples and off-shell triboson VVV(->jj) samples could overlap).

- **Which cross-sections to use?** Whenever possible, best SM predictions should be used, with high-order corrections (QCD and EW) included. The availability of high-order corrections is highly dependent on the physics process. It is generally recommended to use the cross sections directly from AMI database or the [PmgCentralPage](#). There is also a tool available: [PMGCrossSectionTool](#). Some details on the availability of high-order corrections can also be found below in [PmgWeakBosonProcesses#Higher_order_cross_sections](#). If you are able to get high-order corrections from some theorists, or calculate from some specific MC generators, it is also justifiable to apply them in your analysis (assuming it is documented in detail somewhere and has been reviewed).

If you think the Central Page or AMI has an incorrect value for any of your samples, please report it following the instructions [here](#).

- **Which systematic samples to use?** Please refer to the recipes on systematic uncertainties below in [PmgWeakBosonProcesses#Systematic_uncertainties](#).
- **Where the MC samples can be found?** You can either find the detailed DSID for each weak boson process in the subsections below, or find them in the [PmgCentralPage](#). If you find any samples missing, please contact us! It could be that either the samples you are interested in exist but they are not listed, or simply they don't exist.

Validation samples

NEW: List of validation samples and feed back from PA/CP groups can be found at [PmgWeakBosonValidationSamples](#).

V+jets

Summary of sample descriptions

<button class="twistyExpandAll foswikiButton">Expand all</button> <button class="twistyCollapseAll foswikiButton">Collapse all</button>

Z → ee

The samples are divided into two categories, '**Valid**' and '**Obsolete**' samples. The '**Valid**' samples are recommended for ATLAS wide usage, whilst '**Obsolete**' not to be used by analyses and are listed for prosperity reasons only.

Generator	Accuracy	DSID range	Comments
Recommended Samples			
Sherpa 2.2.11	V+0,1,2j@NLO+3,4,5j@LO	700320 - 700322	m(l) > 40 GeV, enhanced in max(HT,pTV), and 2.2.1-like EVNT-to-EVNT heavy flavor filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA - Use and provide feedback
Sherpa 2.2.11	V+0,1,2j@NLO+3,4,5j@LO	700467 - 700469	10 GeV < m(l) < 40 GeV, enhanced in max(HT,pTV), and 2.2.1-like EVNT-to-EVNT heavy flavor filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA - Use and provide feedback
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364114-364127	m(l) > 40 GeV, max(HT,pTV) and B/C sliced.
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364218	pTV [500,1000] GeV and >1000 GeV (latter only MC16). Overlap with max(HT,pTV) slices need to be removed if using together. See pg 3 here
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364204-364209	10 GeV < m(l) < 40 GeV, fewer slices and only BFilter & BVeto
Sherpa 2.2.1	V+0,1j@NLO+2,3,4j@LO	364280	2*m(l) < m(l) < 10 GeV OR 2 GeV < pT(l2) < 5 GeV, only one slice with max(HT,pTV) > 280 GeV
Sherpa 2.2.1	V+0,1j@NLO+2,3,4j@LO	364358 and 36361	2*m(l) < m(l) < 10 GeV OR 2 GeV < pT(l2) < 5 GeV, one slice with 70 GeV < max(HT,pTV) > 140 GeV and one with 140 GeV < max(HT,pTV) < 280 GeV; JIRA
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	366309-366310, 366312-366313, 366315	m(l) > 100 GeV, only low MAXHTPTV slices and select HF filters. AF2. Overlap removal with baseline samples required.
Sherpa 2.2.7	V+0,1j@NLO+2,3,4j@LO	312448 - 312459	pTV,mjj sliced (pTV [100, 140, 220, 280, 500], mjj [0, 500, 1000, CME]), with kT-merging. JIRA
Madgraph+Pythia8 FxFx	V+0-3@NLO	506193 - 506195	m(l) > 40 GeV, enhanced in Ht, with heavy flavour filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA
Madgraph+Pythia8	V+0-4j@LO	363147-363170	m(l)>40 GeV, HT-sliced
Madgraph+Pythia8	V+0-4j@LO	361500 - 361504	m(l)>40 GeV, Np-sliced
Madgraph+Pythia8	V+0-4j@LO	361628 - 361632	10<m(l)<40 GeV, Np-sliced
Powheg+Pythia8	V+0j@NLO	361106	m(l)>60 GeV , JIRA
Powheg+Pythia8	V+0j@NLO	361664	6<m(l)<10 GeV
Powheg+Pythia8	V+0j@NLO	361665	10<m(l)<60 GeV
Powheg+Pythia8	V+0,1@NLO	301000-301018	Mass sliced starting from m(Z) > 120 GeV Overlap needs to be removed with the inclusive samples for coverage of the entire mass range,using the Z boson in the truth record
Obsolete Samples			
Sherpa 2.2.10	V+0,1,2j@NLO+3,4j@LO	700125 - 700137	m(l) > 40 GeV, Enhancement with log(max(HT,pTV)), and 2.2.1-like EVNT-to-EVNT heavy flavor filtering, BFilter, CFilterBveto , CvetoBveto . JIRA .

Z → μμ

The samples are divided into two categories, '**Valid**' and '**Obsolete**' samples. The '**Valid**' samples are recommended for ATLAS wide usage, whilst '**Obsolete**' not to be used by analyses and are listed for prosperity reasons only.

Generator	Accuracy	DSID range	Comments
Recommended Samples			
Sherpa 2.2.11	V+0,1,2j@NLO+3,4,5j@LO	700323-700325	m(l) > 40GeV, Enhanced in max(HT,pTV), and 2.2.1-like EVNT-to-EVNT heavy flavor filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA - Use and provide feedback
Sherpa 2.2.11	V+0,1,2j@NLO+3,4,5j@LO	700470-700472	10 GeV < m(l) < 40 GeV, enhanced in max(HT,pTV), and 2.2.1-like EVNT-to-EVNT heavy flavor filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA - Use and provide feedback
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364100-364113	m(l) > 40 GeV, PDF central values only (no replicas) → transfer relative PDF uncertainty from Z→ee, max(HT,pTV) and B/C sliced
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364216-364217	pTV [500,1000] GeV and >1000 GeV only. Overlap with max(HT,pTV) slices need to be removed if using together. See pg 3 here
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364198-364203	10 GeV < m(l) < 40 GeV, fewer slices and only BFilter & BVeto
Sherpa 2.2.1	V+0,1j@NLO+2,3,4j@LO	364281	2*m(l) < m(l) < 10 GeV OR 2 GeV < pT(l2) < 5 GeV, only one slice with max(HT,pTV) > 280 GeV
Sherpa 2.2.1	V+0,1j@NLO+2,3,4j@LO	364359 and 364362	2*m(l) < m(l) < 10 GeV OR 2 GeV < pT(l2) < 5 GeV, one slice with 70 GeV < max(HT,pTV) > 140 GeV and one with 140 GeV < max(HT,pTV) < 280 GeV; JIRA
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	366300-366301, 366303-366304, 366306	m(l) > 100 GeV, only low MAXHTPTV slices and select HF filters. AF2. Overlap removal with baseline samples required.
Sherpa 2.2.7	V+0,1j@NLO+2,3,4j@LO	312460-312471	pTV,mjj sliced (pTV [100, 140, 220, 280, 500], mjj [0, 500, 1000, CME]), with kT-merging. JIRA
Madgraph+Pythia8 FxFx	V+0-3@NLO	506196 - 506198	m(l) > 40 GeV, enhanced in Ht, with heavy flavour filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA
Madgraph+Pythia8	V+0-4j@LO	363123-363146	m(l)>40 GeV, HT-sliced
Madgraph+Pythia8	V+0-4j@LO	361505-361509	m(l)>40 GeV, Np-sliced
Madgraph+Pythia8	V+0-4j@LO	361633-361637	10<m(l)<40 GeV, Np-sliced
Powheg+Pythia8	V+0j@NLO	361107	m(l)>60 GeV, JIRA
Powheg+Pythia8	V+0j@NLO	361666	6<m(l)<10 GeV, JIRA
Powheg+Pythia8	V+0j@NLO	361667	10<m(l)<60 GeV, JIRA
Powheg+Pythia8	V+0,1@NLO	301020-301038	Mass sliced starting from m(Z) > 120 GeV Overlap needs to be removed with the inclusive samples for coverage of the entire mass range,using the Z boson in the truth record
Obsolete Samples			
Sherpa 2.2.10	V+0,1,2j@NLO+3,4j@LO	700128 - 700130	m(l) > 40 GeV, Enhancement with log(max(HT,pTV)), and 2.2.1-like EVNT-to-EVNT heavy flavor filtering, BFilter, CFilterBveto , CvetoBveto . JIRA .

Z → tt

The samples are divided into two categories, '**Valid**' and '**Obsolete**' samples. The '**Valid**' samples are recommended for ATLAS wide usage, whilst '**Obsolete**' not to be used by analyses and are listed for prosperity reasons only.

Generator	Accuracy	DSID range	Comments
Recommended Samples			
Sherpa 2.2.11	V+0,1,2j@NLO+3,4,5j@LO	700326- 700334	m(l) > 40GeV, enhanced in max(HT,pTV), split according to tau decay modes [LL,LH,HH], and 2.2.1-like EVNT-to-EVNT heavy flavor filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA - Use and provide feedback
Sherpa 2.2.11	V+0,1,2j@NLO+3,4,5j@LO	700473- 700481	10 GeV < m(l) < 40 GeV, enhanced in max(HT,pTV), split according to decay modes [LL,LH,HH], and 2.2.1-like EVNT-to-EVNT heavy flavor filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA - Use and provide feedback
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364128- 364142	m(l) > 40 GeV, max(HT,pTV) and B/C sliced
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364220- 364221	pTV [500,1000] GeV and >1000 GeV only. Overlap with max(HT,pTV) slices need to be removed if using together. See pg 3 here
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364210- 364215	10 GeV < m(l) < 40 GeV, fewer slices and only BFilter & BVeto
Sherpa 2.2.1	V+0,1j@NLO+2,3,4j@LO	364282	2*m(l) < m(l) < 10 GeV OR 2 GeV < pT(l2) < 5 GeV, only one slice with max(HT,pTV) > 280 GeV
Sherpa 2.2.1	V+0,1j@NLO+2,3,4j@LO	364360 and 36363	2*m(l) < m(l) < 10 GeV OR 2 GeV < pT(l2) < 5 GeV, one slice with 70 GeV < max(HT,pTV) > 140 GeV and one with 140 GeV < max(HT,pTV) < 280 GeV; JIRA
Sherpa 2.2.7	V+0,1j@NLO+2,3,4j@LO	312472- 312483	pTV,mjj sliced (pTV [100, 140, 220, 280, 500], mjj [0, 500, 1000, CME]), with kT-merging. JIRA
Madgraph+Pythia8 FxFx	V+0-3@NLO	512198- 512200	m(l) > 40 GeV, enhanced in Ht, with heavy flavour filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA
Madgraph+Pythia8	V+0-4j@LO	361510- 361514	m(l)>40 GeV, Np-sliced
Madgraph+Pythia8	V+0-4j@LO	361638- 361642	10<m(l)<40 GeV, Np-sliced
Powheg+Pythia8	V+0j@NLO	361108	m(l)>60 GeV JIRA , preferred e-tag: e3601
Powheg+Pythia8	V+0j@NLO	361668	6<m(l)<10 GeV
Powheg+Pythia8	V+0j@NLO	361669	10<m(l)<60 GeV
Powheg+Pythia8	V+0,1@NLO	301040- 301058	Mass sliced starting from m(Z) > 120 GeV Overlap needs to be removed with the inclusive samples for coverage of the entire mass range,using the Z boson in the truth record
Obsolete Samples			
Sherpa 2.2.10	V+0,1,2j@NLO+3,4j@LO	700131- 700139	m(l) > 40 GeV, Enhancement with log(max(HT,pTV)), sliced into LepLep , LepHad , HadHad for tau decays, and each decay is then transformed with heavy flavor filters as above.

Z → vv

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Generator	Accuracy	DSID range	Comments
Recommended Samples			
Sherpa 2.2.11	V+0,1,2j@NLO+3,4,5j@LO	700335 - 700337	m(vv) > 2GeV, enhanced in pTV with a pTV >70GeV integration bound, and 2.2.1-like EVNT-to-EVNT heavy flavor filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA - Use and provide feedback
Sherpa 2.2.10	V+0,1,2j@NLO+3,4j@LO	700140-700142	m(vv) > 2 GeV, Enhancement with log(pTV), and heavy flavor filtered. JIRA .
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364143-364155	max(HT,pTV) and B/C sliced
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364222-364223	pTV [500,1000] GeV and >1000 GeV only (former only MC16). Overlap with max(HT,pTV) slices need to be removed if using together. See pg 3 here
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	366000 - 366008	pTV and Mjj sliced. EVNT only.
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	366009 - 366035	pTV and Mjj sliced and B filtered (only MC16d), see JIRA . They share the inclusive EVNT produced above (366000-366008), and are produced by EVNT-to-EVNT filtering. Mjj slicing done for PTV100_140 and PTV140_280 only. For PTV > 500 GeV , no mjj nor b/c filter applied, and one should use the inclusive samples (364222-364223).
Sherpa 2.2.7	V+0,1j@NLO+2,3,4j@LO	312484 - 312495	pTV,mjj sliced (pTV [100, 140, 220, 280, 500], mjj [0, 500, 1000, CME]), with kT-merging. JIRA
Madgraph+Pythia8 FxFx	V+0-3@NLO	509758 - 509760	pTV > 70 GeV, m(vv) > 5 GeV, enhanced in Ht, with heavy flavour filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA
Madgraph+Pythia8	V+0-4j@LO	363195-363218	m(l) $\bar{l}$ >40 GeV, HT-sliced
Obsolete Samples			
Sherpa 2.2.10	V+0,1,2j@NLO+3,4j@LO	700140-2	m(vv) > 2 GeV, Enhancement with log(pTV), and heavy flavor filtered. JIRA .

W $\rightarrow$ $\mu\nu$

The samples are divided into two categories, 'Valid' and 'Obsolete' samples. The 'Valid' samples are recommended for ATLAS wide usage, whilst 'Obsolete' not to be used by analyses and are listed for prosperity reasons only.

Generator	Accuracy	DSID range	Comments
Recommended Samples			
Sherpa 2.2.11	V+0,1,2j@NLO+3,4,5j@LO	700341 - 700343	m(l $\bar{l}$) > 40GeV, Enhanced in max(HT,pTV), and 2.2.1-like EVNT-to-EVNT heavy flavor filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA - Use and provide feedback
Sherpa 2.2.10	V+0,1,2j@NLO+3,4j@LO	700245	mW>120 GeV ; enhancement with log(mW) up to 5 TeV Overlap needs to be removed with inclusive samples. JIRA
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364156-364169	max(HT,pTV) and B/C sliced
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364224-364225	pTV [500,1000] GeV and >1000 GeV only. Overlap with max(HT,pTV) slices need to be removed if using together. See pg 3 here
Sherpa 2.2.7	V+0,1j@NLO+2,3,4j@LO	312496 - 312507	pTV,mjj sliced (pTV [100, 140, 220, 280, 500], mjj [0, 500, 1000, CME]), with kT-merging. JIRA
Madgraph+Pythia8 FxFx	V+0-3@NLO	508982 - 508984	Enhanced in Ht, with heavy flavour filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA
Madgraph+Pythia8	V+0-4j@LO	363624-363647	HT-sliced
Madgraph+Pythia8	V+0-4j@LO	361525 - 361529	Np-sliced
Powheg+Pythia8	V+0j@NLO	361101, 361104	W+, W-
Powheg+Pythia8	V+0j@NLO	301100-301138	Mass sliced starting from m(W) > 120 GeV Overlap needs to be removed with the inclusive samples for coverage of the entire mass range,using the W boson in the truth record
Obsolete Samples			
Sherpa 2.2.10	V+0,1,2j@NLO+3,4j@LO	700146 - 700148	m(lv) > 2 GeV, Enhancement with log(max(HT,pTV)), and heavy flavor filtered. JIRA .

W → ev

The samples are divided into two categories, '**Valid**' and '**Obsolete**' samples. The '**Valid**' samples are recommended for ATLAS wide usage, whilst '**Obsolete**' not to be used by analyses and are listed for prosperity reasons only.

Generator	Accuracy	DSID range	Comments
Valid Samples			
Sherpa 2.2.11	V+0,1,2j@NLO+3,4,5j@LO	700338 - 700340	m(l) > 40 GeV, Enhanced in max(HT,pTV), and 2.2.1-like EVNT-to-EVNT heavy flavor filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA - Use and provide feedback
Sherpa 2.2.10	V+0,1,2j@NLO+3,4j@LO	700246	mW>120 GeV ; enhancement with log(mW) up to 5 TeV Overlap needs to be removed with inclusive samples. JIRA
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364157- 364183	max(HT,pTV) and B/C sliced
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364226- 364227	pTV [500,1000] GeV and >1000 GeV only. Overlap with max(HT,pTV) slices need to be removed if using together. See pg 3 here
Sherpa 2.2.7	V+0,1j@NLO+2,3,4j@LO	312508 - 312519	pTV,mjj sliced (pTV [100, 140, 220, 280, 500], mjj [0, 500, 1000, CME]), with kT-merging. JIRA
Madgraph+Pythia8 FxFx	V+0-3@NLO	508979 - 508981	Enhanced in Ht, with heavy flavour filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA
Madgraph+Pythia8	V+0-4j@LO	363600- 363623	HT-sliced
Madgraph+Pythia8	V+0-4j@LO	361520 - 361524	Np-sliced
Powheg+Pythia8	V+0j@NLO	361100, 361103	W+, W-
Powheg+Pythia8	V+0j@NLO	301060- 301098	Mass sliced starting from m(W) > 120 GeV Overlap needs to be removed with the inclusive samples for coverage of the entire mass range,using the W boson in the truth record
Obsolete Samples			
Sherpa 2.2.10	V+0,1,2j@NLO+3,4j@LO	700143 - 700145	m(lv) > 2 GeV, Enhancement with log(max(HT,pTV)), and heavy flavor filtered. JIRA .

W → tv

The samples are divided into two categories, '**Valid**' and '**Obsolete**' samples. The '**Valid**' samples are recommended for ATLAS wide usage, whilst '**Obsolete**' not to be used by analyses and are listed for prosperity reasons only.

Generator	Accuracy	DSID range	Comments
Valid Samples			
Sherpa 2.2.11	V+0,1,2j@NLO+3,4,5j@LO	700344 - 700349	m(l) > 40GeV, Enhanced in max(HT,pTV), split according to tau decay modes [LL,LH,HH], and 2.2.1-like EVNT-to-EVNT heavy flavor filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA - Use and provide feedback
Sherpa 2.2.10	V+0,1,2j@NLO+3,4j@LO	700247	mW>120 GeV ; enhancement with log(mW) up to 5 TeV Overlap needs to be removed with inclusive samples. JIRA
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364184-364197	max(HT,pTV) and B/C sliced
Sherpa 2.2.1	V+0,1,2j@NLO+3,4j@LO	364228-364229	pTV [500,1000] GeV and >1000 GeV only. Overlap with max(HT,pTV) slices need to be removed if using together. See pg 3 here
Sherpa 2.2.7	V+0,1j@NLO+2,3,4j@LO	312520 - 312531	pTV,mjj sliced (pTV [100, 140, 220, 280, 500], mjj [0, 500, 1000, CME]), with kT-merging. JIRA
Madgraph+Pythia8 FxFx	V+0-3@NLO	509751 - 509756	Enhanced in Ht, with leptonic and hadronic decay mode filtering ('L', 'H'), and heavy flavour filtering (BFilter, BVetoCFilter, BVetoCVeto) - JIRA
Madgraph+Pythia8	V+0-4j@LO	363648-363671	HT-sliced
Madgraph+Pythia8	V+0-4j@LO	361530 - 361534	Np-sliced
Powheg+Pythia8	V+0j@NLO	361102, 361105	W+, W-
Powheg+Pythia8	V+0j@NLO	301140-301178	Mass sliced starting from m(W) > 120 GeV Overlap needs to be removed with the inclusive samples for coverage of the entire mass range,using the W boson in the truth record
Obsolete Samples			
Sherpa 2.2.10	V+0,1,2j@NLO+3,4j@LO	700149 - 700151	

V → qq

Generator	Accuracy	DSID range	Comments
Sherpa 2.2.11	V+0,1j@NLO + 2,3,4,5@LO	700598-700599	Recommended baseline samples, with pT(Z) > 200 GeV , enhanced in pTV, and split into Z(qq)/Z(bb) decay modes - overlaps with 700440-700441 - JIRA
		700439-700441	Recommended baseline samples, with pTW(Z) > 200(100) GeV , enhanced in pTV, and Z(qq/bb)/W(qq) decays modes available - JIRA
Sherpa 2.2.8	V+0,1 @NLO , 2,3,4j@LO	700040-700042	pTV>200 GeV , enhanced in log(pTV) up to 1 TeV , JIRA
Sherpa 2.2.8	V+0,1 @NLO , 2,3,4j@LO	700043-700045	Systematic variation to above Sherpa 2.2.8 samples with Lund hadronization model JIRA
Sherpa 2.2.8	V+0,1 @NLO , 2,3,4j@LO	700080-700091	Systematic variation to above Sherpa 2.2.8 samples with resummation (qsf) and merging scale (ckkw) variations JIRA
Sherpa 2.2.5	V+0,1,2,3,4j@LO	364375-7 Z(qq)	pTV sliced, JIRA
		364378-80 W(qq)	pTV sliced, JIRA
Sherpa 2.1	V+0,1,2,3,4j@LO	304307-304309 W(qq)	pTV sliced, JIRA
		304707-304709 Z(qq)	
Herwig++	V+0,1j@LO	304673-304677, W(qq)	pTV sliced, 65<m(V)<115 GeV , JIRA
		304678-304682, Z(qq)	
Pythia8		304623-304627, 304824, W(qq)	pTV sliced, JIRA
		304628-304632, 304825, Z(qq)	

Electroweak Vjj (VBF)

Generator	Accuracy	DSID range	Comments
Sherpa 2.2.1	Vjj+0,1,2j@LO	308092-308096	Vjj samples including VBF and V strahlung diagrams. These do not include semileptonic VV diagrams! These samples do not overlap with the baseline V+jets samples so they can be combined.
Powheg+Pythia8	Vjj+0j@NLO	344265-6	Electroweak Z(->ee/mumu)jj. This setup uses the global-recoil scheme which isn't supported for VBF/VBS setups. A dipole setup is under study https://its.cern.ch/jira/browse/ATLMCPROD-8756
Powheg+Pythia8	Vjj+0j@NLO	364655-8	Electroweak W(->enu/muvu)jj. This setup uses the global-recoil scheme which isn't supported for VBF/VBS setups. A dipole setup is under study https://its.cern.ch/jira/browse/ATLMCPROD-8756
Herwig 7.1.3	Vjj+0j@NLO	363233-39	Eelectroweak Vjj. JIRA . Caveat: the muon channels suffer from an issue that leads to displaced vertices. Please see discussions at the JIRA.
Herwig 7.2.0	Vjj+0j@NLO	830006-7	Eelectroweak W(munu)jj, Z(mumu)jj. JIRA . Intermediate samples to fix the muon displaced vertices.

Diboson (VV, Vgamma)

The following section will summarise the available samples centrally produced by the Physics Modelling Group for diboson (VV & Vy) processes. Below is a new single comprehensive sample list built using the new [centralpage](#) machinery. It is currently work in progress however it will soon replace the sub-sections below.

Description	DSID range
Baseline	
Sherpa228_Vy	Details 700011-700017
Sherpa_VV_EWK	Details 364283-364287
Sherpa_VV_QCD	Details 345043-345044,345705-345706,345718,345723,363356,363358-363360,364250,364253-364255,364288-364290,364302-364305
Sherpa_2211_Zqqy	No DSIDs found for hashtag combination
Sherpa_2211_Zbby	No DSIDs found for hashtag combination
Sherpa_2211_Zlly	No DSIDs found for hashtag combination
Sherpa_2211_Zlly	No DSIDs found for hashtag combination
Alternative	
Sherpa_VV_QCD	No DSIDs found for hashtag combination
PowhegPythia_VV_QCD	Details 342556,343232,361601,361603-361604
PowhegPythia_VV_EW	No DSIDs found for hashtag combination
Sherpa228_Vy_pTV90	Details 700018-700024
Sherpa_2211_Zll_gamma_jets_OffShell	No DSIDs found for hashtag combination
Specialised	
Sherpa_VV_QCD	Details 364251-364252,364261
MG5Py8_Wlv_gamma_jets	No DSIDs found for hashtag combination

Fully-leptonic diboson

Generator	Accuracy	DSIDs	Comments
Sherpa 2.2.2	0,1j@NLO + 2,3j@LO	364250, llll	mll(SFOS) > 4 GeV, pT1>5 GeV, pT2>5 GeV; 364254 excludes ssWW (use 364286)
		364253, llvv	
		364254, llvv	
		364255, llvv	
		364288, llll_lowMIIPtComplement	mll>2*ml+250MeV, (pT1>20GeV OR MET>50GeV) AND (pT2<5GeV OR ONE M(SFOS)<4GeV). Can be combined with the complementary samples above to get the full phase space.
		364289, llvv_lowMIIPtComplement	
		364290, llvv_lowMIIPtComplement	
		345666 , llvZZ	Special sample, should not be combined with 364254. This sample requires lepton flavor and neutrino flavor to be different, to suppress WW->llvv. One needs to scale the cross section by a factor of 1.5 to account for all three neutrino flavors for ZZ->llvv.
		345667 , llvWW	Special sample, should not be combined with 364254. This sample requires two different lepton flavors, to suppress ZZ->2l2v. One needs to scale the cross section by a factor of 1.5 to account for all three lepton flavors for WW->llvv.
		345668 , llvInt	Special sample, should not be combined with 364254. This sample requires three specific final states 2e2ve, 2mu2vmu, 2tau2vtau, so that the interference between ZZ->2l2v and WW->2l2v is included.
Sherpa 2.2.1	0,1j@NLO + 2,3j@LO	363494	vvv
PowhegBox +PYTHIA8	0j @ NLO	361600, WW(lvv)	on-the-fly generation, reference
		361601, WZ(lvl)	
		361602, WZ(lvv)	
		361603, ZZ(lll)	
		361604, ZZ(vvl)	
		361605, ZZ(vvv)	
		361602, WZ(lvv)	
		361602, WZ(lvv)	
MG5_aMC@NLO	0,1j@NLO	346790-346791	lll (off-shell), mll > 4 GeV, pt(l) > 1 GeV, JIRA
	0,1j@NLO	364455	lllv, JIRA(mc15)
	0,1,2j@NLO	361292-3, WZ(lvl)	On-shell WZ->lvl with 0,1,2j@NLO using FxFx , JIRA , Extension

Semi-leptonic diboson

Generator	Accuracy	DSIDs	Comments
Sherpa 2.2.1	0,1j@NLO + 2,3j@LO	363355, ZqqZvv	onshell diboson production with factorised decays
		363356, ZqqZll	
		363357, WqqZvv	
		363358, WqqZll	
		363359, WpqqWmlv	
		363360, WplvWmqg	
		363489, WlvZqq	
PowhegBox +PYTHIA8	0j @ NLO	361606, WW(lvqq)	on-the-fly generation, reference
		361607, WZ(qqll)	mll>20 GeV
		361608, WZ(qqvv)	
		361609, WZ(lvqq)	mqq>20 GeV
		361610, ZZ(qqll)	mqq>20 GeV
		361611, ZZ(vvqq)	mqq>20 GeV

Electroweak VVjj

Generator	Accuracy	DSIDs	Comments
Sherpa 2.2.2	0j@LO	364283, llljj_EW6	These samples include triboson VVV diagrams where one of the bosons decays hadronically V->jj, also VH and VBFH; 364285 excludes ssWWjj (use 364287). Note that the llvjj_os_EW6 364285 is buggy and should not be used.
		364284, llvjj_EW6	
	0(,1)j@LO	364287, llvjj_ss_EW6	same-sign ll
		366086-89, [llll, llv, llvjj_Min_N_TChannel	With the triboson contributions REMOVED. See JIRA.
PowhegBox+PYTHIA8	VVjj @ NLO	363995, 363996	ssWWjj EWK, lthe inputs, reference
PowhegBox+PYTHIA8	VVjj @ NLO	363997, 363998	ssWWjj QCD, lthe inputs, reference
MadGraph5+PYTHIA8	VVjj @ LO	364848-364854, (WW/WZ/ZZ)jj_EW6 semileptonic decay	EW VVjj with semileptonic decays, on-shell VV only, JIRA
		361294, 361295, WWjj_EW6	EW W(lv)W(qq)jj with B-filter and B-veto
		364499, llvjj_EW6	EW llvjj, similar to 364284, Sherpa2.2.2 llvjj_EW6. The Sherpa sample 364284 has a b-veto so the tZ contributions are not included. This Madgraph 364499 sample is inclusive i.e also has tZ diagrams included.
		364739-42, llljjEW6	EW WZ(->lll)jj, similar to 364499, but with "SpaceShower:dipoleRecoil = on", see PmgWeakBosonProcesses#Dipole_shower_for_VBS_VBF_proces for more detail. Systematic weight variations included, see JIRA
		500986-8, llvjj_ss_EW6	Same-sign WW(->llv)jj, separate samples for QCD-induced and EW-induced and the interference, JIRA
MadGraph5+H7	VVjj @ LO	500989-91, llvjj_ss_EW6	Same-sign WW(->llv)jj, separate samples for QCD-induced and EW-induced and the interference, JIRA

Some notes:

- Usually Sherpa and MadGraph EW VVjj samples include both VBS VVjj and Triboson VVV(->jj) diagrams, while Powheg and VBFNLO samples don't include VVV.

Loop-induced gg -> VV

Generator	Accuracy	DSIDs	Comments
Sherpa 2.1.1	0,1j@LO	361073,361077	fully offshell 4-lepton production; specialised samples without Higgs contributions or in special phase space regions exist in the same DSID range
Sherpa 2.2.2	0,1j@LO	345705-345727	Special splitting into WW/ZZ/interfering final states or With/No/Only Higgs diagrams as follows (details in https://its.cern.ch/jira/browse/ATLMCPROD-4946):
		(baseline) 345705	gg->llll with m4l<130
		(baseline) 345706	gg->llll with m4l>130
		345707-345713	gg->llll specialised samples[*]
		345714	gg->llvv including all final states
		(baseline) 345715	gg->llvv only interfering (WW/ZZ) final states (e.g. e ev e ve)
		345716-17	gg->llvv only int, specialised samples[*]
		(baseline) 345718	gg->llvv only WW-like final states (e.g. e ve mu vmu)
		345719-22	gg->llvv only WW-like, specialised samples[*]
		(baseline) 345723	gg->llvv only ZZ-like final states (e.g. e e vmu vmu)
		345724-27	gg->llvv only ZZ-like, specialised samples[*]
	0,1j@LO	364302	gg->ZllZqq final states (using factorised on-shell decays)
	0,1j@LO	364303	gg->ZvvZqq final states (using factorised on-shell decays)
	0,1j@LO	364304	gg->WmlvWpqq final states (using factorised on-shell decays)
	0,1j@LO	364305	gg->WplvWmq final states (using factorised on-shell decays)

[*] with Higgs diagram requirement/veto, and partially with 5*Gamma_H

Vgamma

Generator	Accuracy	DSIDs	Comments
Vgamma Sherpa 2.2.2	0,1j@NLO + 2,3j@LO	364500-35 , [ll/v]gamma_pty	Z(->ee/mumu/tautau/nunu)gamma, W(->enu/munu/taunu)gamma, sliced in pTgamma (7, 15, 35, 70, 140, CME). Warning: known issues on photon pt slicing.
Zgamma Sherpa 2.2.2	0,1j@NLO + 2,3j@LO	345887-900 , [ll]gamma_M	Z(->ee/mumu)gamma, sliced in m(Z). JIRA .
Zgamma Madgraph +FxFx	0,1j@NLO	345775-84 , [ll]gamma_HTxxxx_Mjixx	Z(->ee/mumu/tautau/nunu)gamma, sliced in HT, mjj, JIRA(mc15) .
Zgamma Sherpa 2.1	0,1,2,3j@LO	363520-7 , [ee, mumu]gammaPt	Z(->ee/mumu)gamma, sliced in photon pt. JIRA .
Zgamma Sherpa 2.2.4	0,1,2,3j@LO	366140-54 , [ll]gamma_LO_pty	Z(->ee/mumu/tautau)gamma, sliced in photon pt. JIRA .
Vgamma Sherpa 2.2.4	0,1@NLO+2,3j@LO	366100-34 , [ll/v]gamma_pty	Z(->ee/mumu/tautau)gamma, W(->enu/munu/taunu)gamma, sliced in photon pt. JIRA , Aborted.
Vgamma Sherpa 2.2.8	0,1@NLO+2,3j@LO	700011-17 , [ll/v]gamma_pty	Z(->ee/mumu/tautau)gamma, W(->enu/munu/taunu)gamma, pt(y) > 7 GeV , enhanced in log10(max(pT(V), pT(y))). JIRA .
Vgamma Sherpa 2.2.11	0,1j@NLO + 2,3,4j@LO	700398-700404 , [ll,lv]gamma	Z(->ee/mumu/tautau/nunu)+gamma, W(enu,munu,taunu)+gamma, enhanced using max[pT(V), pT(y)] with a minimum pT(y)>7 GeV . Recommended for full Run-2 analyses - JIRA .

Some notes:

- Overlap removal between V+Jets and V+gamma:** There is a dedicated tool developed by Hannes Mildner to deal with the overlap removal. See more details at [VGammaORTool](#), with some additional instructions in the presentation [here](#).
- NLO/LO k-factor for Zgamma:** The SM Zg analysis team derived a k-factor of 1.25 to account for the NLO/LO difference in their ISR region (defined by $m_{\ell\ell} + m_{\ell\ell\gamma} > 182 \text{ GeV}$). Please see <https://cds.cern.ch/record/2644987/files/ATL-COM-PHYS-2018-1493.pdf>, section 4.3 for more detail.

V(-> qq)gamma

Generator	Accuracy	DSIDs	Comments
Z(->qq/bb)gamma Sherpa 2.2.11	0,1j@NLO + 2,3j@LO	700352-700353	Z(->qq/bb)gamma, enhanced in max[pT(Z), pT(y)] (on-shell)
V(->qq)gamma Sherpa 2.1	0j@LO	305435-44 , [W, Z]qqGammaPt	W/Z(->qq)gamma, sliced in pTgamma
Z(->qq)gamma Madgraph	0j@LO	364418-22 , [W, Z]qqgammaNp0123_DP	W/Z(->qq)gamma, sliced in pTgamma

Some notes:

- Overlap removal between V+Jets and V+gamma:** There is a dedicated tool developed by Hannes Mildner to deal with the overlap removal. See more details at [VGammaORTool](#).
- Overlap between V(-> qq)gamma and Gamma+jets?:** The EW corrections to gamma + jets would only be virtual and not include the real emission of V(-> qq). Therefore, in the current samples of V(-> qq)gamma and Gamma+jets, there is no overlap. Only one does the full pp-> qq gamma calculation at NLO EW, it would include the interference effects.

Electroweak Vgammajj

Generator	Accuracy	DSIDs	Comments
Sherpa 2.2.4	0j@LO	363514-19 , [epnu/emunu ...]gjj_EW5	EW W(->enu/munu/taunu)g production, JIRA .
		363510-13 , [ee, mumu, tautau, nunu]gjj_EW5	EW Z(->ee/mumu/tautau/nunu)g production, JIRA .
MadGraph5 +PYTHIA8	0j@LO	361273-75 , Wgamma2jEWK	EW W(->lv)gammajj, pt(gamma)> 130 GeV , with "SpaceShower:dipoleRecoil = on" (see PmgWeakBosonProcesses#Dipole shower for VBS VBF proces for more detail), JIRA .
		364458-60 , Zgamma2jEWK	EW Z(->ee,mumu,nunu)gjj, pt(gamma)> 130 (10) GeV for Z->nunu (ee/mumu), JIRA .
		361290 , Znunugamma2jEWK	EW Z(->vv)gammajj, pt(gamma)> 130 GeV , with "SpaceShower:dipoleRecoil = on". JIRA .
		363266-72 , EW Zg, Wg	pt(gamma)> 10 GeV , JIRA .
MadGraph5 +Herwig7	0j@LO	361291 , Znunugamma2jEWK	EW Z(->vv)gammajj, pt(gamma)> 130 GeV , JIRA .
		361270-2 , Wgamma2jEWK	EW W(->lv)gammajj, similar to 361273-75, JIRA .

Triboson (VVV, VVgamma, Vgammagamma)**Summary of sample descriptions**

Triboson (VVV)

Generator	Accuracy	DSIDs	Comments
Sherpa 2.2.2	0j@NLO + 1,2@LO	363507-9	onshell(!) VVV production with factorised decays, WWZ_3l1v2j_EW6, WZZ_4l2j_EW6, WZZ_3l1v2j_EW6. These should not be combined with 364283-7 to avoid double-counting!
Sherpa 2.2.2	0j@NLO + 1,2@LO	364242-49	onshell(!) VVV production with factorised fully leptonic decays
Sherpa 2.2.2	0j@NLO + 1,2,3j@LO	364336-9	onshell(!) WWW production with factorised decays to l ν l ν l ν . WARNING: Please do not mix with 364242 as the opposite signed samples already include the 3l3v decay channel. Combining the 364336-9 samples will give 2l2vjj + 3l3v decays. Use only 364242 for higher statistics in only the 3l3v channel. JIRA
Sherpa 2.2.1	0,1j@LO	407311-15	includes both on- and off-shell contributions to 6-lepton production (which means all diagrams contributing to pp->6l are included, not only resonant or on-shell pp->3V->6l contributions.) Note that V(->l ν)H(->VV->4l), l=e,mu,tau,neutrinos, contribution is also included.
VBFNLO+PYTHIA8	0j @ LO	342099 342100	WWW, VBFNLO v2.7.0, lhe inputs, includes off-shell contributions

Some notes:

- Comparison between off-shell and on-shell VVV samples can be found at [Chris.G's study](#).

VVgamma

Generator	Accuracy	DSIDs	Comments
Sherpa 2.2.5	0j@NLO + 1,2,3j@LO	366160 , WZy	onshell(!) production with factorised decays, WZ->l ν l ν , JIRA
		366161 , WWy	onshell(!) production with factorised decays, WW->l ν l ν , JIRA
		366162 , ZZy	onshell(!) production with factorised decays, ZZ->4l, JIRA

V+gamma+gamma

There will be some overlap of these samples with the V+gamma(+gamma from QED FSR) samples listed above. This needs to be removed at the analysis level if you are using both types of sample, see [VGammaORTool](#) for details.

Generator	Accuracy	DSIDs	Comments
Sherpa 2.2.10	0j@NLO+1,2j@LO	700195-201	statistics enhanced in max(leading photon pT, myy) up to 500 GeV, see ATLMCPROD-8592 . Recommended for full Run-2 analyses
MadGraph_aMC@NLO+Pythia8	0,1j@NLO (FxFx)	507585-507588 (Zy); 504678-504680 (Wy)	Zy split into myy80
Sherpa 2.2.4	0,1,2j@LO	364860-80	pt(y) and m(yy) slices, see ATLMCPROD-5058
Sherpa 2.2.2	0j@NLO+1,2j@LO	364550-84	see ATLMCPROD-5372
Sherpa 2.1.1	0,1,2j@LO	343311-17	pTgamma>17 && mgammagamma>80
Sherpa 2.1.1	0,1,2j@LO	407022-28 (+ 343244-46)	pTgamma>50 (+ 10<pTgamma<50 for Wyy)

Higgs samples

 **This page is work in progress!**

For Higgs boson production includin gg->H, VBF->H and VH Higgs processes, please refer to: [HiggsMCSampleList](#)

Systematic uncertainties

 **This page is work in progress!**

General recipes

PMG recommendations for the evaluation of systematic uncertainties can be found in [PmgSystematicUncertaintyRecipes](#). For uncertainties implemented as on-the-fly weights, PMG provides the [PMGSystematicsTool](#) for an automatic calculation and combination of the relevant uncertainties in ROOT or Rivet-based analyses.

Process-specific recipes

V+jets

Sherpa 2.2.1

For analyses using the baseline Sherpa 2.2.1 V+jets samples, the following sources of theoretical uncertainty should be considered:

- Scale variations:** 7-point QCD scale variations stored as internal weights, from which the envelope around the nominal can be derived. The scale uncertainty is then given by the maximum shift of the envelope with respect to the nominal.
- Parton Density Function:** The 100 NNPDF replicas can be passed through the LHAPDF tool as it has built-in methods to work out the combined PDF uncertainty, cf. [PdfRecommendations](#). The central PDF values for the NNPDF3.0nnlo (PDF261000 or 303200 for the hessian method), MMHT2014nnlo68cl (PDF25300) and CT14nnlo (PDF13000) PDF sets are included as variations as well as two NNPDF3.0nnlo alpha_s variations (PDF269000 and PDF270000).
- CKKW and QSF variations:** Parameterized V+jets uncertainties ([CDS](#)) are available and cover all scale variations including CKKW and QSF variations. Run [AddZJetsWeights.py](#) script (for both W+Jets and Z+jets) with the input file located at dev/SUSYTools here: [/cvmfs/atlas.cern.ch/repo/sw/database/GroupData/dev/SUSYTools/Vjets_SysParameterization/VJetsWeights.root](#). The instruction can be found [here](#).

Some notes :

- Note: The PDF weights are missing in Sherpa 2.2.1 Zmumu samples.* These were produced before we figured out whether the production system would be able to handle >100 multi-weights. You can transfer the relative uncertainty that you get for Zee to Zmumu at the truth level (in the same fiducial phase-space).
- The collinear phase space is known to be mis-modeled with Sherpa (See [M. Gignac's talk at BGF](#)). If your analysis is sensitive to this phase space, consider quoting additional uncertainty by comparing e.g. [MadGraph](#) vs Sherpa.
- It's a known issue that for Sherpa 2.2.1 V+jets samples, in some (rare) cases the nominal MC weights might be different from the weights corresponding to the trivial scale variation of "muF=1, muR=1" (see the [example](#) from the pt(Z) measurement). This is due to a (very rare) weight component coming from the parton shower that hasn't been propagated through to the variations. When estimating systematic uncertainties, it's **recommended to use the tool [PMGSystematicsTool](#) and [PmgEventWeights](#)**. If it's not possible to use these PMG tools, it's recommended to work out the scale-uncertainty band with respect to the "muF=1, muR=1" variation and transfer the resulting uncertainty onto the prediction based on the nominal weight.

Sherpa 2.2.11

For analyses using the new Sherpa 2.2.11 V+jets samples, the following sources of theoretical uncertainty should be considered:

- Scale variations** 7-point QCD scale variations stored as internal weights in the baseline samples, from which the envelope around the nominal can be worked out. The scale uncertainty is then given by the maximum shift of the envelope with respect to the nominal. However, two versions of this scale variation exist:
 - ME+PS:** Coherent variation of the matrix element and parton shown renormalisation and factorisation QCD scales. (baseline recommendations for general physics analyses.)
 - ME only:** Variation of the matrix element renormalisation and factorisation QCD scales. (For special studies in case any analysis need the ME only impacts.)
- Parton Density Function** The Hessian NNPDF3.0 NNLO uncertainty scheme is to be adopted, which can be accessed via the LHAPDF tool as it has built-in methods to work out the combined PDF uncertainty, cf. [PdfRecommendations](#). The central PDF values for the NNPDF3.0nnlo (PDF261000), MMHT2014nnlo68cl (PDF25300) and CT14nnlo (PDF13000) PDF sets are included as variations as well as two NNPDF3.0nnlo alpha_s variations (PDF269000 and PDF270000).
- CKKW and QSF variations:** Variation of the ME+PS merging/matching parameter (CKKW), and the resummation scale (QSF) are available as dedicated samples - see [ATLMCPROD-9445](#). The recommendation at present is to assess the systematics at EVNT-level via comparison to the baseline Sherpa 2.2.11 datasets.
 - Note:** Parameterisation of the systematic uncertainties using a calibrated likelihood ratio classifier approach (see [CARL-Torch](#)) on the final state kinematic phase space is currently on-going. The parameterised uncertainties using the calibrated neural network will then be stored internally of the MC event weight storage element of the xAOD EDM. This work however is still on-going and is not yet deployed - see [MCML forum](#) for more details.
- Electroweak Virtual Corrections:** Approximate NLO EW corrections are included as on-the-fly weights using the electroweak virtual approximation. The electroweak and QCD components can be combined using an additive, multiplicative, or exponentiated prescription (cf. section 6 of <https://arxiv.org/abs/1412.5157> rel="nofollow noreferrer noopener" target="_blank">arXiv:1412.5157 or Section 3.1 of [ANA-PMGR-2021-001](#)). Recommendations on how to adopt a correction and apply a systematic uncertainty are given [here](#).

Multibosons

Normalisation uncertainty

Important note: NNLO QCD and NLO EWK corrections * : See [here](#) for details. ***In short:** Just applying a one-value k-factor of the order of 1.1 is generally *not* advisable since in high-pT tails QCD and EW corrections are large and of opposite sign. In the more inclusive analyses simple k-factors might be more defensible, but to have a global recommendation we would need NNLO QCD + NLO EW + PS event generation, suitable for all phase spaces probed. Please contact PMG if you wish to apply such corrections and want to make sure they are suitable for your analysis.

Even though for some processes the NNLO QCD and/or NLO EW calculations are available ([examples](#)), please note that the uncertainties associated with the high-order corrections for a given process usually depends on the phase space used for the calculation. For analyses targeting specific phase spaces, for instance, high pT or high m(VV) tails, high jet multiplicities, the normalisation uncertainties in those fiducial phase spaces are usually much larger. The recipes described below should be followed to estimate the normalisation and/or shape uncertainties.

Samples and approach to be used for VV uncertainties if Sherpa is nominal

The following sources of theoretical uncertainties should be considered.

- Scale variations:** QCD scale uncertainties can be estimated from the envelope of the 7-point variations of the factorisation and renormalisation scales as described in [PmgSystematicUncertaintyRecipes](#).
- Parton Density Function** The 100 NNPDF replicas can be passed through the LHAPDF tool as it has built-in methods to work out the combined PDF uncertainty, cf. [PdfRecommendations](#). The central PDF values for the NNPDF3.0nnlo (PDF261000), MMHT2014nnlo68cl (PDF25300) and CT14nnlo (PDF13000) PDF sets are included as variations as well as two NNPDF3.0nnlo alpha_s variations (PDF269000 and PDF270000).
- CKKW and QSF variations:** Variations of the resummation scale (QSF) and the CKKW matching scale have to be estimated using explicit variation samples. These samples are only available for some final states so far, but can be requested by analyses looking at multijet final states. Since VV + >=4 jets are generated from parton shower, analyses sensitive to these kind of multijet final states may wish to consider additional algorithmic parton-shower uncertainties, e.g. due to the choice of dipole recoil scheme. For this reason, samples with different PS recoil scheme (CSS_KIN_SCHEME=1) have been generated, cf. <https://its.cern.ch/jira/browse/ATLMCPROD-4573> (Illl), <https://its.cern.ch/jira/browse/ATLMCPROD-4620> (Illv), <https://its.cern.ch/jira/browse/ATLMCPROD-4942> (Illv). Reminder: Setting

'CSS_KIN_SCHEME=0' (default) corresponds to using the variation recoil scheme proposed in [Hoe09a](#), while 'CSS_KIN_SCHEME=1' enables the original recoil strategy documented in [Sch07a](#).

Caveat: in general, the CKKW/QSF systematic variations should be considered, but these samples are not available for all processes for the moment. Within PMG we are trying to have all of them available in the future.

The impact of recoil-scheme variation is minor, as can be seen in this example (https://twiki.cern.ch/twiki/pub/AtlasProtected/MultibosonFocusGroup/plots_CSSKIN.pdf) from SS+jets SUSY analysis (X. Poveda).

VV Powheg+Pythia8 as nominal

For shape uncertainties, the prescription to estimate them requires the usage of alternative samples or the usage of internal weights were available. The following variations are considered:

- **Scale variations:** for new samples produced with Pythia8.2, use internal weights for variations of renormalization and factorization scales.
- **PDF:** use internal reweighting where available.
- **Tuning variations for AZNLO:** see [MC Tuning twiki](#)
- **Shower uncertainties:** alternative samples using Herwig++ should be used. Note that a possible overlap with the Tuning variations might have to be taken into account (not evaluated yet)

VVV (V=W,Z)

There are two sets of generators available: VBFNLO (+Pythia8) and Sherpa 2.2. See also [TribosonTwiki](#) and checks performed and documented in the [Multi-boson PUB note](#). VBFNLO samples are only available for WWW. Sherpa samples are available in all cases. Comparison of the x-sections for processes available with both generators show a difference of about 20% - equivalent to the VV case. Part of the difference arises from the difference in EWK parameters.

For Sherpa 2.2 samples, the recipe for VV samples described above [PmgWeakBosonProcesses#Samples_and_approach_to_be_used](#) should be followed.

Treatment of Higgs interference : The Sherpa sample contain all (tree-level) diagrams for pp->WWW with 3 electroweak couplings, so that include the Higgsstrahlung diagrams. However, since these samples use factorized decays with on-shell vector bosons, the resonant contribution from those diagrams can not be reached from the 125 GeV Higgs. The VBFNLO sample include all Higgs interference. Hence, the following usage is recommended:

- Sherpa as nominal: WWW Sherpa samples should be ADDED to WH samples (where nominal is Powheg+Pythia8)
- VBFNLO: use WWW sample only.

Direct differences between the two samples will lead to artificially large differences - do it with care (e.g. follow approach in PUB note to remove Higgs-component from VBFNLO).

Some notes

- **Alternative MC samples:** Generator comparison (like Sherpa vs Powheg, [aMC@NLO](#), etc) are in principle always recommended as it's a sensible thing to do in the analysis, but there are some caveats: 1) it is not recommended to blindly compare two generators, without knowing the detailed setup of each MC generator (like comparing a multi-leg merged/matched setup with an inclusive setup). One needs to do this with care. See the discussion in [PmgSystematicUncertaintyRecipes#Including_modelling_systematics](#). 2) in practice alternative samples are not available for all processes, for example the [FxFx](#) alternative samples (with respect to Sherpa) are available only for some certain final states. Within PMG we are trying to have more alternative setups (eg. [FxFx](#)) available in the future (HELP is welcome!).

Known modelling issues

 This page is work in progress!

Mismodelling in collinear V+jet topologies

Topologies where the boson gets close to a jet are not well modelled by the generators. For a systematic study of the mismodelling, see [Matthew's presentation](#). In this configuration, the boson represents the real-emission part of the NLO electroweak correction to QCD multijet production. It is not clear how much of the mismodelling can be attributed to missing higher-order corrections.

Lepton eta asymmetry in Sherpa 2.2.1. V+jets

An asymmetry between the positive and negative part of the lepton-pseudorapidity distribution has been observed in all Sherpa 2.2.1 V+jets samples. The asymmetry has been traced back to NLO-matched parton-shower emissions for non-trivial colour structures, such as V+2-jet NLO matrix elements. The bug only affects the 2-jet matrix elements in this setup and is therefore mainly visible in such phase-space regions, where the effect is at the level of 2%.

The bug has been fixed as of Sherpa release 2.2.4. Currently there is no plan to release a general reweighting procedure for the existing samples, however if this problem causes any particular issues in your analysis, please contact the FG+PMG convenors. Additional details are reported in this [presentation at the PMG plenary](#).

Steps in B-filtered Sherpa 2.2.1 Z samples

Steps in the B-filtered Sherpa 2.2.1 samples have been observed in Z and b-jet p_{T} distributions. These effects are currently under investigation, and have been discussed in a [Focus Group meeting](#).

Normalisation discrepancies due to heavy-flavour-tags

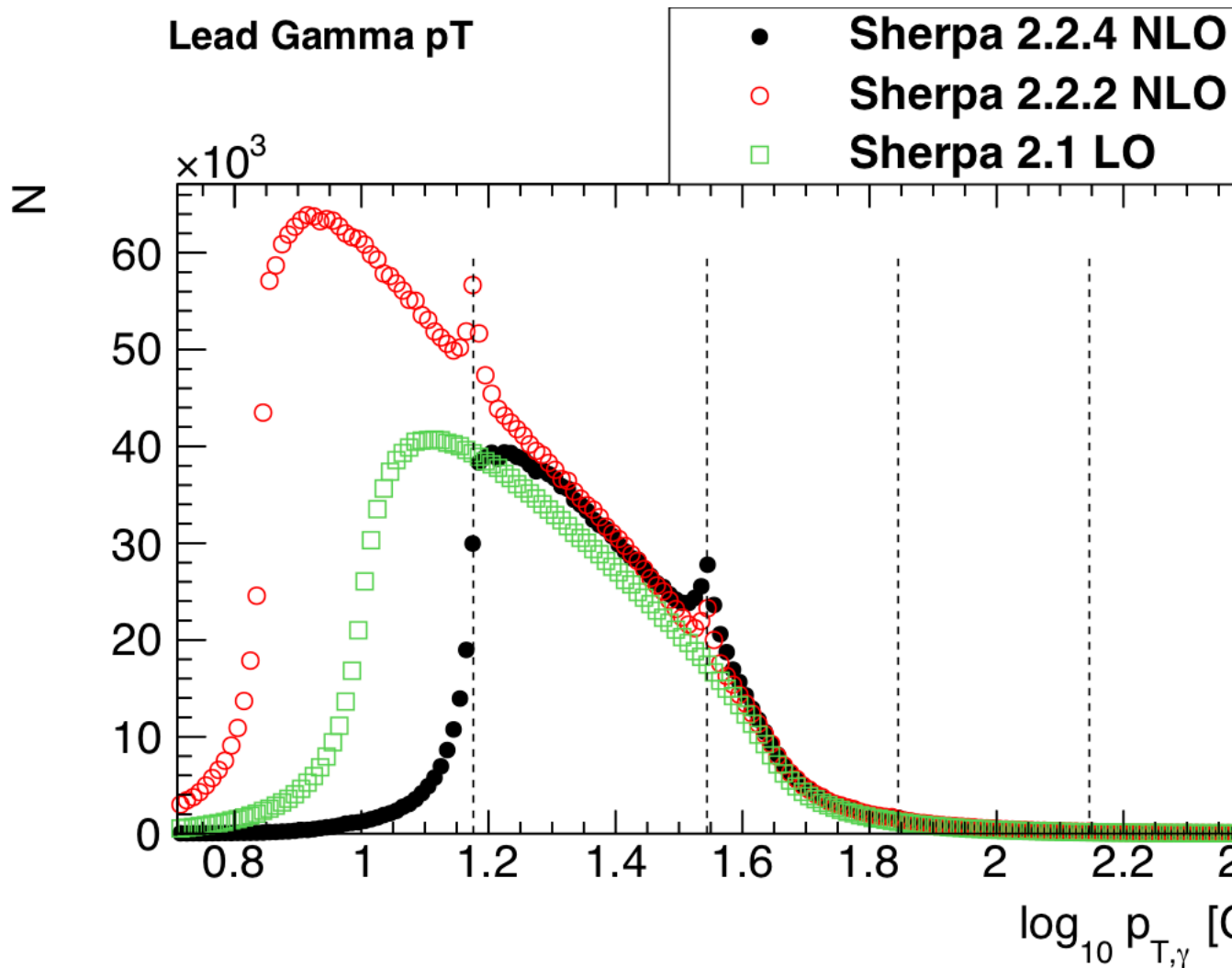
Requiring one or more heavy-flavour tags can lead to normalisation differences on the order of 10-30% between data and simulation. This seems to be the case for all the generators, see e.g. Section 5 in the [boson+jets pub note](#) or the studies collected on this [livepage](#).

Mjj mismodelling in Sherpa

Discrepancies with respect to data and other generators are observed at large dijet invariant masses (beyond a TeV) in $V \rightarrow 2\text{jet}$ events for pretty much all the generators. This has been discussed in two dedicated Focus Group meetings: <https://indico.cern.ch/event/642365/>, and <https://indico.cern.ch/event/645868/>. The effect can also be seen in these [slides](#) as well as the slides from the 13 TeV VBF Z analysis on the [V+jets livepage](#). For Sherpa, this is believed to be due a combination of the shower activity as well as the scale choice for NLO real emissions.

Slicing issues for NLO Sherpa Vgamma samples

Sherpa NLO Vgamma [364501-364535](#) and Vgammagamma [364552-364584](#) samples which have been sliced in photon p_T suffers from a problem when they are stitched together (bumps basically appears in the photon p_T distribution at the boundaries between samples, as it can be seen in the plot below (more details can be found here <https://its.cern.ch/jira/browse/ATLMCPROD-6437>). The stitching problems were mainly seen in the lowest slice, i.e. around the $p_{T\gamma}=15\text{ GeV}$ boundary but can significantly affect the higher p_T slices. **We discourage the use of these Sherpa 2.2.2. NLO Vgamma samples, and instead analyses should use Sherpa 2.2.8 samples.**



Non-optimal color flow setup in Sherpa 2.X VBS/VBF processes

Sherpa VBS samples with versions 2.X suffer from a non-optimal setting of the color flow setup for the parton shower on top of VBS-like scattering processes, leading to an excess of central emissions from the parton shower. For Sherpa CKKW merged VBS samples, this would lead to significantly reduced predicted cross sections (could be as large as 20%). For more detail, please see a presentation by the Sherpa author at MBI 2018, <https://indico.cern.ch/event/700961/contributions/3069895/>; this effect is also studied in the ssWW pub note, see [PmgWeakBosonProcesses#Modelling_of_ssWW](#).

This feature is supposed to be fixed in Sherpa 3.0.

Dipole shower for VBS/VBF processes

Pythia 8 with its default settings produces samples with too much radiation in the central region of rapidity for VBS processes. This tendency can be mitigated by using a new dipole-recoil scheme (<https://arxiv.org/abs/1710.00391>), where only one final-state parton takes the recoil of an emission, rather than the default p_T -ordered shower for ISR emissions, where the recoil of an ISR emission is taken by the whole final state. The effects of changing the recoil scheme are studied in the ssWW pub note, see [PmgWeakBosonProcesses#Modelling_of_ssWW](#).

The dipole-recoil scheme in Pythia 8 is not compatible with the [aMC@NLO](#) matching scheme (although it is compatible with Powheg matching). The dipole shower implementation in Herwig 7 is not compatible with [aMC@NLO](#) matching either.

For LO samples, it's recommended to turn on the dipole shower


```
include("MC15JobOptions/Pythia8_A14_NNPDF23L0_EvtGen_Common.py")
include("MC15JobOptions/Pythia8_MadGraph.py")
genSeq.Pythia8.Commands += [
    "SpaceShower:dipoleRecoil = on"]
```

NLO EW corrections in Sherpa2.2.1X diboson samples

Issue: Applying the NLO EW virtual corrections in Sherpa2.2.1X (2.2.11 onwards up to the current latest Sherpa version 2.2.14) diboson samples affects the overall normalization by -10%.

Expectation: From literature (2005.12128, 2111.13453) we expect the normalization effect to be marginal at low mass (-2 to -3%).

Shape effects: The virtual NLO EW corrections reproduce well the expected impact on the shape of the common diboson observables.

Temporary solution: Apply the NLO EW corrections to obtain the correct shape of the observables, but keep the nominal normalization without the NLO EW corrections. Technically achievable by using the alternative event weight and normalizing the sample to the alternative sum of initial weights– this cancels out the -10% effect. The default procedure for all other samples is to normalize the variations to the nominal sum of initial weights as explained https://twiki.cern.ch/twiki/bin/view/AtlasProtected/PmgSystematicUncertaintyRecipes#On_the_fly_systematic_variations.

Current On-going Weak Boson MC Sample Requests

The following is a list of on-going MC sample requests pertinent to the Weak Boson sub-group, available via JIRA:

Summary	JIRA link	Reporter	Status
Sherpa 2.2.11 Z($\rightarrow$ ee/mumu)y + jets Run-3 MC	ATLMCPROD-9552	Matthew Gignac	To Do
Inclusive Powheg+Pythia8 Z $\rightarrow$ ll Run-3 samples	ATLMCPROD-9551	Matthew Gignac	To Do
Sherpa 2.2.11 V $\rightarrow$ qq samples	ATLMCPROD-9526	Matthew Gignac	To Do
Production of V+0-3jets FxFx MG+Py8 HT2bias samples	ATLMCPROD-9515	Jan Kretzschmar	To Do
Sherpa 2.2.11 W($\rightarrow$ qq')y + jets	ATLMCPROD-9504	Matthew Gignac	To Do
W+jets setups with SMT filter	ATLMCPROD-9473	Marco Vanadia	To Do
Sherpa 2.2.11 WWjj noHiggs/highMjj setup with improved colour flow description	ATLMCPROD-9460	Christian Gutschow	To Do
MG_aMC@NLO+Pythia off-shell Diboson samples	ATLMCPROD-9456	Matthew Gignac	To Do
Sherpa 2.2.11 V+jets CKKW/QSF Variation	ATLMCPROD-9445	Yuzhan Zhao	To Do
EW WW MadGraph +Herwig7 sample request	ATLMCPROD-9443	Robin Hayes	Approved
Extra samples needed for PMG paper	ATLMCPROD-9442	Matthew Gignac	To Do
Updated Powheg+Pythia8 VBF W,Z samples	ATLMCPROD-9440	Jan Kretzschmar	To Do
Sherpa 2.2.10 ZZ(llbb) Reverted Heavy Flavour Modelling Validation Samples	ATLMCPROD-9432	Stephen Jiggins	To Do
Sherpa 2.2.11 low-m(ll) Z+jets validation setups	ATLMCPROD-9419	Aidan Sean Kelly	To Do
Sherpa 2.2.11 VBF setups with improved colour flow	ATLMCPROD-9418	Christian Gutschow	To Do
HWW Offshell signal samples	ATLMCPROD-9408	Bryan Kortman	Approved
Sherpa 2.1 gamma+jets extension for FTAG	ATLMCPROD-9407	Christian Gutschow	To Do
W+jets setups with forced D-meson decays	ATLMCPROD-9405	Christian Gutschow	To Do
MadGraph WVZ samples	ATLMCPROD-9404	Thomas Glyn Hitchings	To Do
Sherpa 2.2.11: ssWWjj (VBS) with improved colour flow	ATLMCPROD-9392	Christian Gutschow	To Do
gg $\rightarrow$ (H*) $\rightarrow$ ZZ $\rightarrow$ 4l sample with upto 2 ME jets matched and merged with parton shower	ATLMCPROD-9386	Jay Ajitbhai Sandesara	Approved
Updated Sherpa 2.2.11 V $\rightarrow$ qq samples	ATLMCPROD-9360	Matthew Gignac	To Do
Systematic WWY samples with off-shell W bosons	ATLMCPROD-9354	Muhammad Alhroob	To Do
Madgraph Zyy NLO samples	ATLMCPROD-9332	Ruchi Gupta	To Do
Sherpa 2.2.4 lgamma at 8 TeV (EVNT only)	ATLMCPROD-9322	Alexey Kharlamov	To Do
Request for new samples with higher statistics	ATLMCPROD-9320	Sandeep Kaur	To Do
Background samples for Hl: diboson processes	ATLMCPROD-9309	Iwona Grabowska-Bold	To Do
V+jets validation setups for CPU/modeling	ATLMCPROD-9306	Matthew Gignac	To Do
Extension for Sherpa 2.2.10 NLO llyy	ATLMCPROD-9297	Daniel James Lewis	To Do
Sherpa 2.2.11 low mass Ztautau samples	ATLMCPROD-9283	Matthew Gignac	To Do
Sherpa 2.2.11 ll+y and lnu+y samples	ATLMCPROD-9282	Matthew Gignac	To Do
WW EW generation with Madgraph+pythia8	ATLMCPROD-9238	Yun-Ju Lu	Approved
Sherpa 2.2.11 Vqqy+jets samples	ATLMCPROD-9220	Matthew Gignac	To Do
Sherpa 2.2.11 V+jets production samples	ATLMCPROD-9214	Matthew Gignac	To Do
Sherpa 2.2.11 0-2j@NLO + 3,4,5j@LO validation samples	ATLMCPROD-9181	Matthew Gignac	To Do
Sherpa 2.2.11 Zee/Wmunu HTprime for unordered histories validation samples	ATLMCPROD-9152	Matthew Gignac	To Do
Sherpa 2.2.11 4FS Z+bb validation samples	ATLMCPROD-9151	Matthew Gignac	To Do
Sherpa 2.2.11 Z+jets macro sliced validation samples	ATLMCPROD-9131	Matthew Gignac	To Do
Sherpa 2.2.11 Zee+jets sliced validation samples	ATLMCPROD-9085	Matthew Gignac	To Do
Sherpa 2.2.11 Shower improvements	ATLMCPROD-9081	Matthew Gignac	To Do
Sherpa 2.2.11 Enhance function validation samples	ATLMCPROD-9073	Matthew Gignac	To Do
Off-Shell H - ZZ ggF samples with dimension-six SMEFT operators	ATLMCPROD-9059	Jay Ajitbhai Sandesara	Approved
Sherpa 2.2.10 pTV2 enhance function validation sample	ATLMCPROD-9047	Matthew Gignac	To Do
Madgraph Wy+jets samples	ATLMCPROD-9038	Heather Russell	To Do
Production of Znuu+3jets FxFx	ATLMCPROD-8989	Francesco Giuli	Approved
Sherpa 2.2.X Vy merging/resummation scale variations	ATLMCPROD-8983	Matthew Gignac	To Do
Production of V+3jets FxFx : towards the final setup	ATLMCPROD-8930	Federico Sforza	Approved

The status of these requests is summarised below:

Practical information

Common text for MC samples description for Papers/CONF notes

PMG is maintaining common tex files to describe commonly used MC samples for SM processes. They can be found at: <https://gitlab.cern.ch/atlas-physics/pmg/documents/references>. Both short and long descriptions are available. You may find them useful when writing papers or CONF notes. If you use these tex files,

please check carefully to make sure they match exactly the samples used in your analysis. Any feedback is welcome!

- Single-boson processes: <https://gitlab.cern.ch/atlas-physics/pmg/documents/references/blob/master/BJ.tex>
- Multiboson processes: <https://gitlab.cern.ch/atlas-physics/pmg/documents/references/blob/master/MB.tex>

Higher order cross sections

Single-boson processes

K-factors for the latest Sherpa 2.2.11+ and Madgraph FxFx models. NNLO cross sections have been calculated using MATRIX. For more details see [PubNotes](#). These should be applied to all V+jets MC samples including the high and low mLL samples in order to have a consistent normalisation at the sample boundaries. For multi-leg generators, the k-factors should be applied for inclusive phase-spaces without any jet selections. After requiring the presences of jets, the generator cross section may be used without the k-factors applied.

CoM energy: 13 TeV

Generator	Process	K-factor
Sherpa 2.2.11+	$Z \rightarrow \ell\ell$	0.95
Sherpa 2.2.11+	$Z \rightarrow \nu\nu$	0.95
Sherpa 2.2.11+	$W^\pm \rightarrow \ell\nu$	0.95
MGFxFxPythia8	$Z \rightarrow \ell\ell$	0.93
MGFxFxPythia8	$Z \rightarrow \nu\nu$	0.93
MGFxFxPythia8	$W^\pm \rightarrow \ell\nu$	0.93
PowhegPythia8	$Z \rightarrow \ell\ell$	1.03
PowhegPythia8	$W^+ \rightarrow \ell\nu$	1.03
PowhegPythia8	$W^- \rightarrow \ell\nu$	1.04

CoM energy: 13.6 TeV

Generator	Process	K-factor
Sherpa 2.2.11+	$Z \rightarrow \ell\ell$	0.94
Sherpa 2.2.11+	$Z \rightarrow \nu\nu$	0.94
Sherpa 2.2.11+	$W^\pm \rightarrow \ell\nu$	0.94
MGFxFxPythia8	$Z \rightarrow \ell\ell$	0.92
MGFxFxPythia8	$Z \rightarrow \nu\nu$	0.92
MGFxFxPythia8	$W^\pm \rightarrow \ell\nu$	0.92
PowhegPythia8	$Z \rightarrow \ell\ell$	1.03
PowhegPythia8	$W^+ \rightarrow \ell\nu$	1.03
PowhegPythia8	$W^- \rightarrow \ell\nu$	1.04

For older versions of models - Sherpa 2.2.2/MadgraphPythia at 13 TeV Inclusive cross sections at NNLO accuracy in the strong coupling constant have been calculated using FEWZ at various centre-of-mass energies. An overview of the available K-factors is given below. A **5% uncertainty** should be applied on the normalisation when these K-factors are used. More details on how these K-factors were obtained can be found on this [twiki](#). First values for comparison with NNLO calculations [Note in preparation](#) are entered and once verified will be added to PMG cross section file.

Generator	Process	K-factor
Sherpa 2.2.1	$Z \rightarrow \ell\ell$	0.9751
Sherpa 2.2.1	$Z \rightarrow \nu\nu$	0.9728
Sherpa 2.2.1	$W^\pm \rightarrow \ell\nu$	0.9702
MadgraphPythia8	$Z \rightarrow \ell\ell$ (Np0-4 slices)	1.232
MadgraphPythia8	$Z \rightarrow \nu\nu$ (Np0-4 slices)	1.2283
MadgraphPythia8	$W^\pm \rightarrow \ell\nu$ (Np0-4 slices)	1.20185
MadgraphPythia8	$Z \rightarrow \ell\ell$ (HT slices)	1.141
MadgraphPythia8	$W \rightarrow \ell\nu$ (HT slices)	1.12
PowhegPythia8	$Z \rightarrow \ell\ell$	1.026
PowhegPythia8	$W^+ \rightarrow \ell\nu$	1.01724
PowhegPythia8	$W^- \rightarrow \ell\nu$	1.035786
AlpgenPythia6	$Z \rightarrow \ell\ell$	1.196
AlpgenPythia6	$W \rightarrow \ell\nu$	1.1109

Multi-boson processes

For multiboson processes, the NLO QCD corrections are known for most processes, and NNLO QCD + NLO EW corrections are also becoming available for main diboson processes. For most Powheg inclusive single-boson/diboson samples, the calculation is already at NLO QCD. For most Sherpa samples, the accuracy is at 0,1j@NLO +2,3j@LO or 0,1j@LO. Since the high-order corrections are sensitive to the phase space definition, and the application to a given MC setup can also be tricky (inclusive or merged/matched setup), it is not feasible to provide them centrally. Some existing study could be found at this [twiki](#). Within PMG we are also planning to provide high-order corrections for as many processes as possible, **help is very welcome and you would be awarded with OTP!**. Feel free to contact us if you are interested.

Overlap with Higgs signal samples

decay / prod	ggH	VBF	V(→qq)H	V(→ll)H
bb/cc/qq	--	--	--	-- (j) (m)
VV→llll a)	gg→llll (b)	llllj EWK (f)	llllj EWK (f)	lllll (k)
VV→llqq a)	-- (c)	-- (g)	-- (g) (m)	llllj EWK (f)
VV→qqqq	-- (d)	-- (g)	-- (g)	-- (g)
μμ/ττ	--	-- (h)	-- (h)	-- (n)
γγ	-- (e)	-- (i)	-- (i)	-- (o)

Footnotes:

- a) l = l⁺l⁻, ν
- b) 100% overlap with 345705+345706+345715+345718+345723. Alternative samples without Higgs diagrams exist, see [PmgWeakBosonProcesses#Loop_induced_gg_VV](#) and ATLMCPROD-4946
- c) gg→V(ll)V(qq) samples are produced with onshell VV diagrams, i.e. they cannot contain an onshell Higgs resonances.
- d) gg→V(qq)V(qq) samples don't exist, but same onshell argument would hold
- e) No overlap with diphoton samples in 364350-54 because they include gg→γγ only as O(alpha^2 alphaS^2) box diagram, no Higgs contribution.
- f) Vector boson scattering samples for llllj (364283 + 364285) fully contain the VBF H→VV→llll and V(qq)H(→VV→llll) and V(ll)H(→VV→llqq) diagrams at LO accuracy.
- g) I'm not aware of offshell llqqq or qqqqq EWK samples, only VVjj and VVV onshell ones, which cannot contain an onshell Higgs resonance.
- h) There are two types of ττ+jets samples: (i) QCD ττ+jets (364128-41 + 364210-15) only include diagrams with O(alpha^2), and by default the tau is set massless in these samples. Both of these exclude H(→ττ)jj VBF diagrams. (ii) VBF ττjj production (308094) would in principle allow 4 EW couplings and thus could include H(→ττ)jj VBF diagrams. But again the tau is set massless so the H(→ττ) vertex is disabled in these. The same holds for μμ.
- i) γγjj EWK samples with 4 EW couplings (like 364306) would not overlap with VBF H(γγ) or V(qq)H(γγ) because the Higgs signals needs 2 additional couplings in the loop-induced H→γγ decay.
- j) lljj EWK samples could overlap here, but since the c/b are treated as massless in the ME, those don't contain diagrams with H(→cc/bb) vertices.
- k) Off-shell "lllll" samples at LO accuracy exist in 407311-15 and include also all H(→VV→llll)V(→ll) diagrams.
- m) Regular QCD V+jets samples are restricted to O(alpha^2), i.e. two EW vertices, and thus cannot contain Higgs diagrams.
- n) Baseline llll samples have massless μ and τ, so no H(→μμ/ττ) diagrams are included.
- o) Baseline llγγ samples are restricted to 4 EW vertices, thus do not include H→γγ one-loop (or effective) vertex.

Treating pathological high-weight events in Sherpa samples

The Sherpa 2.2.1 and 2.2.2 samples contain rare events with very high weight magnitudes ($|w|>100$), in particular for processes with NLO accuracy in high-parton multiplicities.

Technical explanation: These events correspond to phase space points with differential cross sections (weights) larger than the maximal weight estimated during the phase space optimisation. To be completely consistent, during unweighting such events will always be accepted and have to be assigned a remainder weight of $w=\text{weight}/\text{maximum}$ to account for the imperfect MC integration.

These **high** (positive or negative) weights should only happen very rarely, and we basically keep these large weights only to allow for cross-checks that such instabilities don't appear en masse in any particular sample or phase space region. If one such large weight creates a spike in your otherwise well populated distributions, it is fine to ignore it by setting the weight to 1. If you see many such events entering your region of interest it's worth discussing this with PMG to understand whether the MC sample in question would be expected to have problems there.

Note that these large event weights are being taken care of at the generator-level starting from v2.2.4 and so the prescription outlined above should no longer be necessary for Sherpa samples using v2.2.4 or later.

Relevant Rivet routines

Rivet routines are very useful for validation for new MC setups. See [RivetMCValidation](#) for all available and published Rivet routines by ATLAS.

Single boson processes

- Z+jets (13 TeV, 85/pb), [CONF-STD-2015-07](#), [link to Rivet](#).
- Z+jets (13 TeV, 3.2/fb), [STD-2016-01](#), [link to Rivet](#).
- Unfolded Z+jets control region for leptoquark searches (13 TeV): [EXOT-2016-08](#), [link to Rivet](#), contact: Louie Corpe
- Unfolded MET+jets (13 TeV), [STD-2018-55](#), [link to Rivet](#), internal only, by Chris Gutschow.
- Z+HF (13 TeV), [STD-2017-38](#), [link](#), internal only, by Josh [McFayden](#).

Multibon processes

- Inclusive 4l lineshape (13 TeV): [STD-2017-09](#), [link to Rivet](#).

- ZZ->4l (on-shell, 13 TeV): [STDM-2016-15](#), [link to Rivet](#).
- Inclusive WZ->lvll (13 TeV): [STDM-2015-09](#), [link to Rivet](#).

Other useful resources on modeling of weak boson processes

LHC Electroweak Working Group (<https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCEW>) :

- Boson and jets: lhc-ewwg-vjets@cern.ch, [twiki](#)
- Multiboson: lhc-ewwg-multiboson@cern.ch, [twiki](#)

LHC Higgs cross section working group (<https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHXSWG>)

Public results

Pub Notes

Simulation of boson plus jets processes for 13 TeV ATLAS Analyses

PubNote: [single-boson-plus-jets pub note](#). The studies presented in the note have been discussed at the [ATLAS/CMS Monte Carlo workshop](#) in May 2017. The note also documents in detail the setup the current ATLAS Run-2 baseline samples for V+jets and therefore you are encouraged to **reference the pub note** in your papers and notes, IN ADDITION to the generator specific references, which you can find [here](#). The SVN area for the can be found [here](#). The previous V+jets pub note can be found [here](#).

Multi-Boson Simulation for 13 TeV ATLAS Analyses

PubNote: [multi-boson pub note](#). The studies presented in the note have been discussed at the [ATLAS/CMS Monte Carlo workshop](#) in May 2017. The note also documents in detail the setup the current ATLAS Run-2 baseline samples for multi-boson processes and therefore you are encouraged to **reference the pub note** in your papers and notes, IN ADDITION to the generator specific references, which you can find [here](#). The previous pub note (of the now-defunct diboson task force) can be found [here](#).

Modelling of ssWW

PubNote: [ssWW pub note](#). Predicted cross sections for electroweak scattering of two same-sign W bosons in association with two jets (pp->W±W±jj) are compared for a number of Monte Carlo configurations. Fiducial cross sections and kinematic distributions calculated using [Madgraph5](#), [aMC@NLO](#), Powheg and Sherpa are presented.

Major updates:
-- [Main.FrankSiegert](#) - 2019-04-05

Last reviewed by: **Never reviewed**

Vgamma Sherpa 2.2.11	0,1j@NLO + 2,3,4j@LO	700398-700404 [ll,lv]gamma	Z(->ee/mumu/tautau/nunu)+gamma, W(enu,munu,taunu)+gamma, enhanced using max[Ht, pTy] with a minimum pT(y)>7 GeV. Recommended for full Run-2 analyses - JIRA
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I	Attachment	History	Action	Size	Date	Who	Comment
	ptgammalead.pdf	r1	manage	29.2 K	2020-04-09 - 16:23	MatthewGignac	
	ptgammalead.png	r1	manage	153.7 K	2020-04-09 - 16:32	MatthewGignac	

Topic revision: r135 - 2026-08-17 - [WeitaoWang](#)